

Capstone 02- ENGI 4000
Articulate Engineering design problem with parameters and needs of perspective stakeholders

Assignment: Chapter 02	Indicators: DES2	Worth; 5.0% of course grade
Submission Date: September 23, 2025	Group:	
Student Name		
1.		
2.		
3.		
4.		

Late submission penalties:

One day (-1.0 mark)	Two days (-2.0 marks)	Three days (-3.0 marks)	Four days (-4.0 marks)	Five days (-5.0 marks)

Submission will be ignored and marked ZERO if assignment was submitted incorrectly, outside the Blackboard, or submitted in different format than PDF.

Objective

Each group will prepare a mission statement that clearly articulates the engineering design problem, including its parameters and needs of prospective stakeholders for the Capstone project (ENGI 4000 – Capstone I). The report must address the following:

- Conduct market research and review existing solutions
- Describe competitive product features and benefits
- Identify and evaluate development opportunities
- Formulate a clear and concise mission statement

Guidelines

Your submission must be a written report in chapter format and include the following:

- **Cover Page**, including:
 - Chapter title
 - Table of Contents (clearly organized with sections and subsections)
 - Executive Summary (summarizing the key elements of the chapter)
- **Format:** Organize content using clear sections and subsections. Incorporate diagrams, charts, and tables where appropriate, ensuring they include clear titles, labels, and notations.
- **References:** Include all sources used, properly cited in an appropriate and consistent citation format.

Chapter Sections

A. Market Research and Competitive Solutions

Market research is used to assess the viability of a new product or service. In this section, research and include relevant information and data to:

- Identify market segments and key competitors
- Understand consumer needs and expectations to guide the product design process
- List and compare competitive products and their relevant features
- Highlight your team's intended competitive advantages over other companies in the industry and explain how this supports product differentiation

B. Development of Opportunities

This section should explore, and document potential opportunities related to your project. It must include:

- A project charter as a starting point
- A list of opportunities generated by your team (list minimum of 12 opportunities; if fewer, consult your supervisor),

Use a combination of text, charts, sketches, annotated pictures, or other visual tools to present your ideas clearly and concisely.

C. Mission Statement

This section must present a clear and brief mission statement that includes at least the following elements:

- A brief (one-sentence) description of the product
- The benefit proposition (value the product provides)
- Key business goals
- The target market(s)
- Assumptions, risks, and constraints guiding the development effort
- Stakeholders (individuals, groups, or organizations impacted by the project)

Evaluation Rubric (40 marks – equivalent to 5.0% of course grade)

Section	Marks	Criteria
Cover page & Report Quality	5	Proportional layout; includes title, table of contents, and executive summary; clarity of writing; effective use of visuals; proper references
Market research and available solutions	5	Clear approach to fulfill all listed requirements, including the use of charts, schematics, pictures, or other appropriate visual aids to express ideas.
Development of opportunities	18	Strong and well-illustrated presentation of at least 12 opportunities (or as otherwise approved by the supervisor).
Mission statement	12	Includes all six required components shown above (description, benefit proposition, business goals, target market, assumptions/risks/constraints, stakeholders)